

Project Report: Production ML Inference Service - Python Developer

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Family:	Python Developer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

FastAPI service serving a scikit-learn model with batch prediction, caching, OpenTelemetry, and k6 load tests.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Production ML Inference Service** is a advanced-level end-to-end project for Python Developer. **Duration:** 4-6 weeks. **FastAPI service serving a scikit-learn model with batch prediction, caching, OpenTelemetry, and k6 load tests.** **Tools:** FastAPI, sklearn, Redis, k6.

Code Reference:

Objective: Load and version ML model with joblib
Objective: Batch and single prediction endpoints
Objective: Redis result cache
Objective: OpenTelemetry traces and k6 load test

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Python Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Python Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Model [Advanced]

Chapter focus: Phase: Model** *(Level: Advanced)

Train and serialize model. Complete these tasks: sklearn pipeline; MLflow log. Deliverable for this phase: model.pkl v1.

Code Reference:

- [] sklearn pipeline
- [] MLflow log

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Model is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the model.pkl v1 to the teacher.

Chapter 3: Phase: API [Advanced]

Chapter focus: Phase: API** *(Level: Advanced)

Prediction endpoints. Complete these tasks: /predict; /predict/batch. Deliverable for this phase: Latency benchmarks.

Code Reference:

- [] /predict
- [] /predict/batch

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase API is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Latency benchmarks to the teacher.

Chapter 4: Phase: Cache [Advanced]

Chapter focus: Phase: Cache *(Level: Advanced)**

Redis for repeated inputs. Complete these tasks: Cache key hash; TTL config. Deliverable for this phase: Cache hit metrics.

Code Reference:

- [] Cache key hash
- [] TTL config

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Cache is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Cache hit metrics to the teacher.

Chapter 5: Phase: Load Test [Advanced]

Chapter focus: Phase: Load Test *(Level: Advanced)**

k6 script 1000 RPS. Complete these tasks: p95 latency; Error rate. Deliverable for this phase: k6 HTML report.

Code Reference:

- [] p95 latency
- [] Error rate

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Load Test is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the k6 HTML report to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review** *(Level: Advanced)

Submit all final deliverables: Model artifact; k6 report; Grafana/traces screenshot; API docs. Skills demonstrated: ML serving, Caching, Observability. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Model artifact
Deliverable: k6 report
Deliverable: Grafana/traces screenshot
Deliverable: API docs
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Model artifact and passes the rubric.